

Sets to Real Analysis: A Thorough Reconstruction

Contents

Contents	3
1 Introduction to Set Theory	5
1.1 Foundations and Framework	5
1.2 Basic Set Axioms	8
1.3 Set-Coded Binary Relations	17
1.4 Set-Coded Functions	23
Appendices	31

Chapter 1

Introduction to Set Theory

1.1. Foundations and Framework

Logic

A **proposition** Q is anything that can be evaluated as true or false. If P, Q are propositions and P implies Q , we write $P \implies Q$. If P is implied by Q , we write $P \impliedby Q$. If P is true if and only if Q is true, we write $P \iff Q$. We write $\neg P$ to mean the negation of P , $P \wedge Q$ to mean P and Q , and $P \vee Q$ to mean P or Q (inclusively). The usual laws of logic will be assumed without explicating them further.

A **predicate** P is anything that takes in an object x and outputs a proposition $P(x)$. For example, if $P(x)$ means, “ x went to the store,” then P refers to “went to the store.” In this sense, P refers to something that is either true or false of any given object. In addition to predicates that take in a single object, we allow the following: A predicate P may take objects x, y to form a proposition $P(x, y)$ or take objects x, y, z to form a proposition $P(x, y, z)$, depending on what P is precisely. For example, if $P(x, y)$ means, “ x is on top of y ,” then P refers to “is on top of.”

When a discussion takes place, we delineate a **domain of discourse**. This refers to the class of everything that is under consideration in a discussion. For example, when we’re talking about which animals make good pets, the domain of discourse might be limited to domesticated animals like cats, dogs, and rabbits instead of wild animals like tigers or whales. When we say, “all people have been interviewed,” the domain of discourse might be job applicants for a specific position rather than all people in general. By writing $\forall x P(x)$, we mean: For all objects x , we have that $P(x)$. By writing $\exists x P(x)$, we mean: There exists an object x such that $P(x)$. Here $\forall$ is a **universal quantifier** and $\exists$ is an **existential quantifier**. In a statement like $\forall x P(x)$, the variable x is implicitly ranging over some domain like the set of natural numbers or a group of people.

When we deal solely with propositions without quantifiers, we are dealing with **propo-**

sitional logic. When we deal with predicates and quantifiers that range over the objects of the domain of discourse, we are dealing with **first-order** or **predicate logic**. Most of this text will implicitly operate within informal predicate logic. The English language will be used throughout, with logical notation appearing at times only as a shorthand.

Equality

The most basic and prevalent relation in mathematics is the **equality relation** $=$. In mathematics, we use symbols and strings of symbols to designate objects and concepts in a given domain of discourse. If two symbols, say a and b , refer to the same object, we use equality to signify this by writing $a = b$. If the symbols a and b refer to different objects, we signify this by writing $a \neq b$. The following characterization of equality is sufficient for the mathematics we will be undertaking:

- (1) **LAW OF IDENTITY:** For any object x in a given domain of discourse, $x = x$.
- (2) **INDISCERNIBILITY OF IDENTICALS (SCHEMA):** If P is a predicate, then for any objects x, y in a given domain of discourse, if $x = y$ then $P(x) \iff P(y)$.

The **indiscernibility of identicals** is also called the **substitution property**. In the way we formulated it, the indiscernibility of identicals is a **schema** rather than a single statement. This means it represents multiple statements, one for each predicate P . When we later formulate the axioms of set theory, we will encounter **axiom schemas**. An axiom schema is simply a way of representing multiple axioms, one for each predicate.

We will be flexible with our notation in obvious ways. For example, by writing $x = y = z$, we mean that $x = y$ and $y = z$ hold simultaneously. From properties (1) and (2) above, we may prove other properties of equality.

Proposition 1.1. *For any objects a, b, c in a given domain of discourse, we have the following:*

- (a) **SYMMETRY:** *If $a = b$, then $b = a$.*
- (b) **TRANSITIVITY:** *If $a = b$ and $b = c$, then $a = c$.*

Proof. Suppose $a = b$. Let $P(x)$ be the predicate $x = a$. By the law of identity, $P(a)$ holds true. By the substitution property and by the hypothesis that $a = b$, it follows $P(b)$ holds true, which stands for $b = a$.

Suppose $a = b$ and $b = c$. Let $Q(x)$ be the predicate $a = x$. By hypothesis, $Q(b)$ holds true. By the substitution property and by our assumption that $b = c$, it follows that $Q(c)$ holds true, which stands for $a = c$. ■

Constants, Functions, and Relations

In addition to connectives $\wedge, \vee, \implies, \impliedby, \iff$, quantifiers $\exists, \forall$, equality $=$, and parentheses, we will have three kinds of non-logical symbols. A **constant symbol** c is

a symbol that refers to a fixed existing object in the domain of discourse. A **(domain guarded) function symbol** f with **domain condition** D is a symbol that takes in a term t (anything that may refer to an existing object in the domain of discourse) for which a proof of $D(t)$ exists and returns new term $f(t)$ that refers to an existing object in the domain of discourse. A **binary relation symbol** R is a symbol that takes in two terms s, t and returns a proposition $R(s, t)$; we often write $R(s, t)$ as $s R t$.

The constant, function, and relation symbols we are discussing here are syntactic objects outside of the mathematical domain of discourse. If f is a function symbol with domain condition D , we only consider $f(t)$ well-formed if a proof of $D(t)$ exists (in the relevant context) where t is a well-formed term. A function f such that $f(t)$ is defined for all terms t is possible if the domain condition is trivial like $D(t) \equiv t = t$.

Unlike in treatments of a formal theory where the language is declared once and never changes, we will extend our language as our text unfolds. Each extended theory must be conservative over the original theory, meaning that any statement in the old language that can be proven in the new theory must be provable in the old theory. This way, everything entailed and not entailed by our original axioms is preserved. Consequently, each extension can be done only under certain conditions. Each extension takes one of three forms:

- **INTRODUCING CONSTANT SYMBOLS.** Constants refer to existing mathematical objects. If we show

$$\exists x P(x)$$

for some predicate P , then we may introduce a new symbol c with a new axiom

$$P(c).$$

Note that because uniqueness is not established, if d satisfies P , it does not necessarily follow that $c = d$. Oftentimes, however, we may show

$$\exists x (P(x) \wedge \forall y (P(y) \implies y = x))$$

for some predicate P , which states that there exists a unique element satisfying P . We may abbreviate this as $\exists!x P(x)$. In such a case, we may introduce a new symbol c with a new axiom

$$P(c) \wedge \forall y (P(y) \implies y = c).$$

- **INTRODUCING (DOMAIN GUARDED) FUNCTION SYMBOLS.** Function symbols are used to transform terms into other terms. If we show

$$\forall x (D(x) \implies \exists!y P(x, y))$$

for some one-place predicate D and two-place predicate P , then given x such that $D(x)$ holds we want to be able to pick out the unique y such that $P(x, y)$ holds. We do this by introducing a new symbol f with domain condition D and a new axiom

$$\forall x (D(x) \implies P(x, f(x))).$$

Function symbols with the trivial domain condition $D(x) \equiv x = x$ may be reframed as function symbols without domain conditions. Sometimes we will also have D as a two-place predicate and P as a three-place predicate so that given x, y for which $D(x, y)$ holds, we may form $f(x, y)$ so that $P(x, y, f(x, y))$ holds.

- **INTRODUCING BINARY RELATION SYMBOLS.** Binary relation symbols take two terms and return a proposition. In general, for any two-place predicate P , we may add a binary relation symbol R with a new axiom

$$\forall x \forall y (P(x, y) \iff x R y).$$

The conditions before adding new symbols are necessary, because their usage presupposes existence. For example, if our initial axioms prove $\neg \exists x P(x)$, then adding a new constant symbol c such that $P(c)$ introduces a contradiction: We would be able to prove $\neg \exists x P(x)$ and $\exists x P(x)$ simultaneously, breaking not only conservativity over the original theory but consistency as well.

Language and Notation

A **character** is taken to be any symbol in a language. More generally, it can be any atomic piece of a language that can't be broken down further. A **string** is any sequence of characters written together consecutively. We include the **empty string**, which consists of no characters, as a string. In our framework, strings and characters will be linguistic devices sitting outside of our object-level mathematics.

Just like how strings denote mathematical objects, we may sometimes introduce meta-variables and meta-constants to denote strings. For example, we may let $\underline{S}$ denote the string 00517, which is a numeral distinct from the number 00517. Equality $=$ is reserved for object-level terms, while $\equiv$ will be a meta-relation used to denote equality of propositions, predicates, and strings. So going to our numeral vs number example, we have $00517 = 517$ but $00517 \not\equiv 517$, because the latter is a statement about uninterpreted strings.

1.2. Basic Set Axioms

The Conception of Set Theory

Set theory provides powerful tools to organize mathematical concepts. Additionally, set theory is capable of subsuming and encoding a large amount of mathematics, as we will see in our text. The theory we will be working with shall be **Zermelo–Fraenkel set theory with the axiom of choice**, which is abbreviated as **ZFC**. The domain of ZFC consists of only one sort, which is the **set**, so all elements of ZFC are sets which themselves contain other sets and so on.

Intuitively, the universe of sets may be viewed as being generated in stages, beginning with the empty set $\emptyset$. Since $\emptyset$ exists, the set $\{\emptyset\}$ containing only $\emptyset$ may be formed; given these we may form sets such as $\{\{\emptyset\}\}$ and $\{\emptyset, \{\emptyset\}\}$, and so on. Continuing this process through both finite and infinite stages yields the cumulative hierarchy of sets. All steps are specified by the axioms of ZFC.

We will operate with two layers. One layer will involve specifying the structure of mathematical concepts and the other layer will involve implementing the concept within set theory. For example, we might define an ordered pair (a, b) to be an object o whose only relevant information is the objects a and b and the order in which they occur. This is sufficient as a definition in our first layer, but because everything is a set in the second layer, we will provide a set-theoretic realization of this type of object; an example we will later see is the Kuratowski implementation in which $o = \{\{a\}, \{a, b\}\}$. We may specify a theory of a number system axiomatically and then work with that given theory, but in addition to this we will show how the described number system is realized within pure set theory (we will provide a set-theoretic **model** for the given **theory**). The motivation for the latter part is a matter of demonstrating relative consistency. For many theories (such as Peano arithmetic), we cannot establish their consistency in an absolute sense, but we can establish relative consistency by providing set-theoretic realizations of what the theories are describing. If we are reasonably confident that ZFC is consistent, then an implementation of some number system within set theory means we can be reasonably confident the given theory is consistent as well.

Our philosophy will be along the lines of **structuralism**, which is the view that mathematical objects are defined by the roles they play within a structure, rather than by some intrinsic nature they possess independently. Although every mathematical object discussed in this text will be realized as a set, our interest will be in the properties and relationships of mathematical objects rather than in what those objects are. A strong case for structuralism can be made by pointing out that a given concept can be realized in set theory in more than one way and that there are alternative competing foundations of mathematics that are distinct from set theory as well.

Sets, Membership, and Subset Relations

In ZFC, any mention of element or object is taken as synonymous with set. The formal language of ZFC consists of the logical symbols and one additional relation symbol $\in$. However, as we explained above, we will not adhere to this language and we will introduce constant terms, (domain guarded) function symbols, and relation symbols as we see fit. The $\in$ relation represents **membership**, so if a, b are sets and $a \in b$ then we say a is a **member** or **element** of b . If it is not the case that $a \in b$ then we write $a \notin b$.

Let us now provide two axioms.

(ZFC1) AXIOM OF THE EMPTY SET. There exists a set $\emptyset$ such that $\forall x (x \notin \emptyset)$.

(ZFC2) AXIOM OF EXTENSIONALITY. For any A, B , if $\forall x (x \in A \iff x \in B)$, then $A = B$.

The axiom of the empty set asserts the existence of an **empty set**, defined as any set $\emptyset$ satisfying $\forall x (x \notin \emptyset)$.

Proposition 1.2. *The empty set is unique.*

Proof. Suppose $\emptyset, \emptyset'$ have the property that $x \notin \emptyset$ and $x \notin \emptyset'$ for all x . Then for all x , the equivalence $x \in \emptyset \iff x \in \emptyset'$ is vacuously true, so $\emptyset = \emptyset'$ by (ZFC2). ■

We will universally denote the empty set by $\emptyset$ or $\{\}$. Since it is unique, there is no ambiguity as to what these symbols refer to.

If $A = B$, then by the substitution property it is necessarily the case that $\forall x (x \in A \iff x \in B)$, which means A and B have exactly the same elements. However, the converse does not necessarily follow without the axiom of extensionality. The axiom of extensionality asserts that sets are defined by nothing other than what they contain, so if A and B contain exactly the same elements, then $A = B$. (We could imagine perhaps in an alternative version of set theory, sets come in different colors, so A and B would have the same elements but A would be a red set and B would be a green set, thus rendering $A \neq B$. The axiom of extensionality ensures this does not happen.)

Definition. Suppose A and B are sets. If every element of A is an element of B , we say A is a **subset** of B and write $A \subseteq B$. Equivalently, we say B is a **superset** of A and write $B \supseteq A$. If A is a subset of B but not equal to B , we say A is a **proper subset** of B and write $A \subset B$. Equivalently, we say B is a **proper superset** of A and write $B \supset A$.

Proposition 1.3. *For any set A , we have $\emptyset \subseteq A$ and $A \subseteq A$.*

Proof. For any x , the implication $x \in \emptyset \implies x \in A$ is vacuously true, so $\emptyset \subseteq A$. For any x , the implication $x \in A \implies x \in A$ is tautologically true, so $A \subseteq A$. ■

Suppose A and B are sets. If $A = B$, then by the substitution property of equality, $x \in A \iff x \in B$ for all x , so the subset relations $A \subseteq B$ and $B \subseteq A$ both hold. We leave it to the reader to show the converse: If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

We will often abbreviate certain logical sentences in obvious ways. For example, given a set A , instead of writing

$$\forall x (x \in A \implies P(x)) \quad \text{and} \quad \exists x (x \in A \wedge P(x)),$$

we write

$$\forall x \in A : P(x) \quad \text{and} \quad \exists x \in A : P(x)$$

and variants thereof.

Construction Axioms

The following axioms describe how to build new sets out of existing objects. We will use these axioms to define various set operations.

(ZFC3) AXIOM OF PAIRING. Given sets a, b , there exists a set

$$B = \{a, b\}$$

for which $x \in B$ if and only if $x = a$ or $x = b$. Intuitively, B is the set consisting of elements a and b only, and we call this the **unordered pairing** of a and b .

(ZFC4) AXIOM OF UNION. Given a set A , there exists a set

$$B = \bigcup A$$

such that $x \in B$ if and only if $x \in y$ for some $y \in A$. Intuitively, B is the set obtained by taking all members of A and then putting the elements of those members into a single set, and we call this the **union** of elements of A .

(ZFC5) AXIOM SCHEMA OF SEPARATION. Given sets A, c and a (possibly two-place) predicate P , there exists a set

$$B = \{x \in A \mid P(x, c)\}$$

such that $x \in B$ if and only if $x \in A$ and $P(x, c)$. Intuitively, B is the subset of A consisting of elements that have the property P (with parameter c), and we call this the **subset** of A of elements satisfying predicate P (with parameter c).

(ZFC6) AXIOM OF POWER SET. Given a set A , there exists a set

$$B = \mathcal{P}(A)$$

such that for all X we have $X \subseteq A \iff X \in B$. Intuitively, B is the set of all subsets of A , and we call this the **power set** of A .

The axiom schema of separation also goes by the names **specification**, **subsets**, and **restricted comprehension**. The construction axioms assert existence of various sets. Let us go over some basic uniqueness propositions.

Proposition 1.4. *We have the following:*

- (a) *Given sets a, b , the set B asserted by (ZFC3) is unique.*
- (b) *Given a set A , the set B asserted by (ZFC4) is unique.*
- (c) *Given sets A, c and a predicate P , the set B asserted by (ZFC5) is unique.*
- (d) *Given a set A , the set B asserted by (ZFC6) is unique.*

Proof. Given sets a, b , suppose B, B' both satisfy the description given by (ZFC3). Then $x \in B$ if and only if $x = a \vee x = b$ if and only if $x = b \vee x = a$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given a set A , suppose B, B' both satisfy the description given by (ZFC4). Then $x \in B$ if and only if $\exists y (y \in A \wedge x \in y)$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given sets A, c and a predicate P , suppose B, B' both satisfy the description given by (ZFC5). Then $x \in B$ if and only if $x \in A \wedge P(x, c)$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given a set A , suppose B, B' both satisfy the description given by (ZFC6). Then $X \in B$ if and only if $X \subseteq A$ if and only if $X \in B'$. By the axiom of extensionality (ZFC2), $B = B'$. ■

Now that uniqueness is established, we may use the notation invoked in (ZFC3) to (ZFC6) universally. Given that

$$\forall a \forall b \exists! B Q(a, b, B) \quad \text{where} \quad Q(a, b, B) \equiv \forall x (x \in B \iff x = a \vee x = b),$$

we introduce a function symbol Pair such that $\text{Pair}(a, b)$ is B in (ZFC3). After that, we rewrite $\text{Pair}(a, b)$ as $\{a, b\}$ for readability. Given a predicate P , we have

$$\forall A \forall c \exists! B Q(A, c, B) \quad \text{where} \quad Q(A, c, B) \equiv \forall x (x \in B \iff x \in A \wedge P(x, c)),$$

so we may introduce a function symbol Sep_P such that $\text{Sep}_P(A, c)$ is B in (ZFC5). After that, we rewrite $\text{Sep}_P(A, c)$ in **set-builder notation**: $\text{Sep}_P(A, c) = \{x \in A \mid P(x, c)\}$. For (ZFC4) and (ZFC6), we introduce function symbols $\bigcup$ and $\mathcal{P}$ accordingly.

Let us give an interesting application of the axioms we have so far. This famous result concerns the non-existence of a certain set in ZFC set theory.

Proposition 1.5 (Russell's paradox). *The set of all sets does not exist.*

Proof. We proceed by proof by contradiction. Suppose such a set A exists. Let $P(x) \equiv x \notin x$ and apply separation to get the subset $B = \{x \in A \mid x \notin x\}$. If $B \in B$, then the definition of B forces $B \notin B$. If $B \notin B$, then the definition of B forces $B \in B$. In all cases, we obtain a contradiction. Since the existence of B is entailed by the existence of A , our supposition that A exists must be false. ■

Using the construction axioms we introduced, we may start to define new kinds of set operations.

Definition (Binary union). Let A and B be sets. We denote the **binary union of A and B** by

$$A \cup B = \bigcup \{A, B\}.$$

Definition (Binary intersection). We define the **binary intersection of A and B** to be the set

$$A \cap B = \{x \in A \mid x \in B\}.$$

Definition (Set difference). We define the **set difference of A relative to B** , also referred to as the **relative complement of B in A** , to be the set

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

Let us use the axioms of set theory to establish that all three set operations form valid sets out of existing sets.

Proposition 1.6. *For any sets A and B , the sets $A \cup B$, $A \cap B$, and $A \setminus B$ exist and are unique.*

Proof. By the axiom of pairing (ZFC3), there exists the set $\{A, B\}$. By the axiom of union (ZFC4), there exists a set U such that $x \in U$ if and only if $x \in Y$ for some $Y \in \{A, B\}$. The condition $Y \in \{A, B\}$ is precisely equivalent to $Y = A$ or $Y = B$, so then the statement $x \in Y$ is simply $x \in A$ or $x \in B$. This shows U is precisely the set consisting of all elements x such that $x \in A$ or $x \in B$. Thus, U is a binary union of A and B , so $A \cup B$ exists.

To obtain the intersection $A \cap B$, apply the axiom schema of separation (ZFC5) to A with the predicate $P(x) \equiv x \in B$. To obtain the set difference $A \setminus B$, apply the axiom schema of separation (ZFC5) to A with the predicate $Q(x) \equiv x \notin B$.

Uniqueness of all three sets given A and B is left to the reader to prove. ■

If $A \cap B = \emptyset$, we say A and B are **disjoint** or equivalently they have an **empty intersection**. Otherwise, we say A and B **intersect** or equivalently they have a **nonempty intersection**.

Given objects a, b , the order in which we write the elements in $\{a, b\}$ does not matter, because one can show $\{a, b\} = \{b, a\}$; it is the set containing exactly the objects a and b in no particular order. To see this, simply note that $x \in \{a, b\}$ if and only if $x = a \vee x = b$ if and only if $x = b \vee x = a$ if and only if $x \in \{b, a\}$.

Given an element a , we may use the axiom of pairing (ZFC3) to form the set $\{a, a\}$. Since $x \in \{a, a\}$ if and only if $x = a \vee x = a$ if and only if $x = a$, this set contains only the element a . If A is a set such that $\forall x (x \in A \iff x = a)$, then we call A a **singleton**, and the axiom of extensionality (ZFC2) implies $A = \{a, a\}$, which is unique with respect to a . For any a , the singleton containing a is written as $\{a\}$.

Given two singletons $\{a\}$ and $\{b\}$, we may form the union $\{a\} \cup \{b\}$ and get back the pairing $\{a, b\}$ of a and b . We may write $\{a, b\} \cup \{c\}$ as $\{a, b, c\}$. A string $\underline{S}$ lists out all elements of a set if it is either of the form $\{\underline{a}\}$ or $\{\underline{T}, \underline{a}\}$ where $\underline{a}$ is a string denoting an element and $\underline{T}$ lists out all elements of another set. If $\underline{S}$ lists out all elements of a set, then the set denoted by the string $\underline{S} \cup \{\underline{a}\}$ may be also denoted by the string $\underline{S}, \underline{a}$, which lists out all elements of the set. For example, if

$$\underline{S} \equiv \{a, b, c\}$$

then

$$\underline{S} \equiv \{a, b, c\}$$

and then $\{a, b, c\} \cup \{d\}$ is rewritten as $\{a, b, c, d\}$.

Example 1.7 (Union). Suppose

$$A = \{\{a, b, c\}, \{b, c, d\}, \{a, e\}\}.$$

Then the union of elements of A is

$$\bigcup A = \{a, b, c, d, e\}.$$

Example 1.8 (Separation). Suppose i, j, k are distinct elements and

$$A = \{a, b, c, d, e\} \quad \text{where} \quad a = \{i, j\}, \quad b = \{i\}, \quad c = \{j\}, \quad d = \{i, j, k\}, \quad e = \{j, k\}.$$

Let $P(x) \equiv i \in x$ and

$$Q(x) \equiv \exists y \exists z (y \neq z \wedge y \in x \wedge z \in x \wedge \forall w (w \in x \implies (w = y \vee w = z))),$$

so $P(x)$ says x contains element i and $Q(x)$ says x contains exactly two elements. By applying separation, we obtain sets

$$\{x \in A \mid P(x)\} = \{a, b, d\} \quad \text{and} \quad \{x \in A \mid Q(x)\} = \{a, e\}.$$

Example 1.9 (Power set). Let $A = \{a, b, c\}$ with a, b, c distinct from one another. Then the power set of A is

$$\mathcal{P}(A) = \{\{a, b, c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a\}, \{b\}, \{c\}, \emptyset\}.$$

Let us provide some examples showcasing binary set operations.

Example 1.10. Let $A = \{a, b, c\}$, $B = \{j, k\}$, and $C = \{b, c, k\}$ where a, b, c, j, k are distinct elements. Then

$$\begin{aligned} A \cup B &= \{a, b, c, j, k\}, & A \cup C &= \{a, b, c, k\}, \\ A \cap B &= \{\}, & A \cap C &= \{b, c\}, \\ A \setminus B &= \{a, b, c\}, & A \setminus C &= \{a\}. \end{aligned}$$

The binary union operation can be considered as a special case of the general union operator from the axiom of union (ZFC4). In a similar manner, the binary intersection operation can be considered as a special case of a more general intersection operator defined below.

Definition (Intersections). Given a nonempty set A , we define the **intersection** of elements of A as

$$\bigcap A = \{x \in \bigcup A \mid \forall y (y \in A \implies x \in y)\}.$$

This is the set of elements x such that $x \in y$ for all $y \in A$.

If A is nonempty, the existence of $\bigcap A$ follows from the axiom of union (ZFC4) and the axiom schema of separation (ZFC5). If A is empty, we leave $\bigcap A$ undefined, so $\bigcap$ is a domain guarded function symbol with domain condition $D(x) \equiv x \neq \emptyset$. (There is nothing wrong with applying union followed by separation in the case where A is empty, but it is conventional to leave the intersection undefined. The thinking is that we want to preserve the property that $A \subseteq B$ implies $\bigcap A \supseteq \bigcap B$ for all applicable sets A, B . If we had $\bigcap \emptyset$ were defined such that the subset property held, then the fact that $\emptyset \subseteq A$ for all sets A would imply $\bigcap \emptyset$ contains all sets. However, such a set does not exist by Russell's paradox.)

Kuratowski–Cartesian Products

Recall that the order in which we write the elements in the pair $\{a, b\}$ does not matter. Before we define a particular type of set-theoretic construction of an **ordered pair**, let us give a characterization of what we want. Given any two elements a, b we want to be able to form an object $o = \text{OrdPr}(a, b)$ with two operations π_L and π_R that recover the original elements: $\pi_L(o) = a$ and $\pi_R(o) = b$. We also want that if o, o' are two such objects such that $\pi_L(o) = \pi_L(o')$ and $\pi_R(o) = \pi_R(o')$, then $o = o'$. This way, $o = \text{OrdPr}(a, b)$ with π_L and π_R is the object whose information content is the first element a and the second element b .

Let us proceed with a set-theoretic construction. Given sets a, b , repeated applications of the axiom of pairing (ZFC3) allow us to form the set $\{\{a\}, \{a, b\}\}$. We call this a **Kuratowski ordered pair** and denote it as $\langle a, b \rangle$. We can get back elements a and b from $\langle a, b \rangle$ as follows. For the first element, we find

$$\bigcup \bigcap \langle a, b \rangle = \bigcup \bigcap \{\{a\}, \{a, b\}\} = \bigcup \{a\} = a.$$

For the second element, we claim

$$\bigcup \left\{ x \in \bigcup \langle a, b \rangle \mid P(x, \langle a, b \rangle) \right\} = b \quad \text{where} \quad P(x, c) \equiv \bigcup c \neq \bigcap c \implies x \notin \bigcap c.$$

To see this, note that if $a = b$, then $\bigcup \langle a, b \rangle = \bigcap \langle a, b \rangle$ so $P(x, \langle a, b \rangle)$ is true for all x , and the set in the set-builder notation is $\{x \in \{b\} \mid P(x, \langle a, b \rangle)\} = \{b\}$, and thus the left-hand side evaluates to b . If $a \neq b$, then $\bigcup \langle a, b \rangle \neq \bigcap \langle a, b \rangle$ so $P(x, \langle a, b \rangle)$ is true only for $x \notin \bigcap \langle a, b \rangle = \{a\}$, i.e. $x = b$, so the set in the set-builder notation is $\{x \in \{a, b\} \mid x = b\} = \{b\}$ and thus the left-hand side evaluates to b . Consequently,

$$\langle a, b \rangle = \langle a', b' \rangle \iff a = a' \text{ and } b = b'. \quad (1.1)$$

If $a \neq b$ then $\langle a, b \rangle \neq \langle b, a \rangle$, so the order in which we write a and b here matters.

Definition. Given sets A, B , we define the **binary Kuratowski–Cartesian product** to be the set

$$A \times_K B = \{z \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid z = \langle x, y \rangle \text{ for some } x \in A \text{ and } y \in B\}.$$

In other words, $A \times_K B$ is the set of all Kuratowski ordered pairs $\langle x, y \rangle$ where $x \in A$ and $y \in B$.

The existence of $A \times_K B$ given sets A, B follows straightforwardly from the existence of $A \cup B$, the axiom of power set (ZFC6), and the axiom schema of separation (ZFC5). If A, B are nonempty, there exist elements $a \in A$ and $b \in B$, so $\langle a, b \rangle \in A \times_K B$ and $A \times_K B$ is nonempty.

Replacement and Foundation Axioms

The following axioms are meant to further specify what kinds of sets we are allowed to have and what kinds of sets we are not allowed to have.

(ZFC7) AXIOM SCHEMA OF REPLACEMENT. Given a three-place predicate P such that

$$\forall x \forall c \exists! y P(x, y, c), \quad (1.2)$$

and sets A, c , there exists a set B such that

$$\forall x (x \in A \implies \exists y (y \in B \wedge P(x, y, c))) \quad \text{and} \quad \forall y (y \in B \implies \exists x (x \in A \wedge P(x, y, c))).$$

(ZFC8) AXIOM OF FOUNDATION/REGULARITY. If A is a nonempty set, there exists a set $x \in A$ such that $x \cap A = \emptyset$.

Any predicate P satisfying (1.2) is called a **functional predicate**.

Proposition 1.11. *Given sets A, c and a functional predicate P , the set B asserted by (ZFC7) is unique.*

Proof. Given sets A, c and functional predicate P , suppose B, B' both satisfy the description given by (ZFC7). Let b be any element of B , so there exists $a \in A$ such that $P(a, b, c)$. Then there exists an element $b' \in B'$ such that $P(a, b', c)$. Since P is a functional predicate, $b = b'$ and so $b \in B'$. This shows $B \subseteq B'$. By the same reasoning, $B' \subseteq B$, so by the axiom of extensionality (ZFC2), $B = B'$. ■

Given a functional predicate P and sets A, c , the existence and uniqueness of the set given by replacement may be written as $B = \text{Rep}_P(A, c)$, or in set-builder notation,

$$B = \{\mathcal{F}_P(x, c) \mid x \in A\}$$

where $\mathcal{F}_P$ is the function corresponding to the functional predicate P .

Proposition 1.12. *For all sets A , $A \notin A$.*

Proof. Let A be any set. By the axiom of pairing (ZFC3), we may form the set $\{A\}$. Then the axiom of foundation (ZFC8) implies there exists $x \in \{A\}$ for which $x \cap \{A\} = \emptyset$. In this case, $x = A$ so $A \cap \{A\} = \emptyset$. If $A \in A$, then $A \in A \cap \{A\}$, which contradicts the previous sentence. Thus, $A \notin A$. ■

1.3. Set-Coded Binary Relations

Basic Definitions

We introduced equality $=$ as a logical relation and membership $\in$ as a primitive relation. We also introduced subset relations $\subseteq, \supseteq, \subset, \supset$, which act as shorthands for predicates that we could in principle write out without those subset symbols. Now we want to introduce other kinds of relations, but we want to treat them as objects of set theory itself. For this reason, we will introduce new relations by encoding them as sets.

Definition. A **(set-coded) binary relation** R from A to B is a set of the form

$$R = \langle G_R, \langle A, B \rangle \rangle \quad \text{where} \quad G_R \subseteq A \times_K B.$$

If $\langle a, b \rangle \in G_R$, then we say a **is related to b by relation R** , and we write $a R b$. If $\langle a, b \rangle \notin G_R$, then we say a **is not related to b by relation R** , and we write $a \not R b$.

If R is a relation from A to itself, we say R is a relation on A . Most relations we will come across will be on one set. Note that it matters which element appears first and which appears second in the ordered pair $\langle a, b \rangle$, even in the case $B = A$, so we might have $a R b$ even though $b \not R a$.

Example 1.13. Let $A = \{a, b, c, d\}$ where a, b, c, d are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b \rangle \vee x = \langle b, c \rangle \vee x = \langle c, d \rangle \vee x = \langle b, a \rangle \vee x = \langle c, b \rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_\sim = \{x \in A \times_K A \mid P(x)\}$, and we form $\sim = \langle G_\sim, \langle A, A \rangle \rangle$. Since $G_\sim \subseteq A \times_K A$, this is a relation. We may list all the relationships by $\sim$ as follows:

$a \not\sim a$	$a \sim b$	$a \not\sim c$	$a \not\sim d$
$b \sim a$	$b \not\sim b$	$b \sim c$	$b \not\sim d$
$c \not\sim a$	$c \sim b$	$c \not\sim c$	$c \sim d$
$d \not\sim a$	$d \not\sim b$	$d \not\sim c$	$d \not\sim d$

Total Order Relations

Here we introduce two closely related types of relations.

Definition. A **strict total order relation** on A is a relation $<$ with the following properties for all $a, b, c \in A$:

(T1) **IRREFLEXIVITY:** $a \not< a$,

- (T2) ASYMMETRY: if $a < b$ then $b \not< a$,
 (T3) TRANSITIVITY: if $a < b$ and $b < c$, then $a < c$,
 (T4) TRICHOTOMY: $a < b$, $a = b$, or $b < a$.

A **weak total order relation** on A is a relation $\leq$ with the following properties for all $a, b, c \in A$:

- (T1') REFLEXIVITY: $a \leq a$,
 (T2') ANTISYMMETRY: if $a \leq b$ and $b \leq a$, then $a = b$,
 (T3') TRANSITIVITY: if $a \leq b$ and $b \leq c$, then $a \leq c$,
 (T4') COMPARABILITY: $a \leq b$ or $b \leq a$.

Given $<$ we may define a corresponding $\leq$ and vice-versa by declaring $\forall x \forall y (x \leq y \iff x < y \vee x = y)$ and $\forall x \forall y (x < y \iff x \leq y \wedge x \neq y)$. We also refer to total order relations as **linear order relations**.

If A has a strict or weak total order relation, then we say A is a **totally ordered set** or **linearly ordered set** and introduce it as a pair $\langle A, < \rangle$. If A has a strict $<$ and a corresponding weak $\leq$, then the following notation and terminology applies:

- $a < b \iff a$ is less than $b \iff b > a \iff b$ is greater than a .
- $a \leq b \iff a$ is less than or equal to $b \iff b \geq a \iff b$ is greater than or equal to a .

We say A is **orderwise dense** if for each $a, b \in A$ with $a < b$, there exists another element $x \in A$ such that $a < x < b$.

Example 1.14. Let $A = \{a, b, c, d\}$ where a, b, c, d are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b \rangle \vee x = \langle a, c \rangle \vee x = \langle a, d \rangle \vee x = \langle b, c \rangle \vee x = \langle b, d \rangle \vee x = \langle c, d \rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_< = \{x \in A \times_K A \mid P(x)\}$, and we form $< = \langle G_<, \langle A, A \rangle \rangle$. Since $G_< \subseteq A \times_K A$, this is a relation. We may list out all the comparisons $<$ as follows:

$a \not< a$	$a < b$	$a < c$	$a < d$
$b \not< a$	$b \not< b$	$b < c$	$b < d$
$c \not< a$	$c \not< b$	$c \not< c$	$c < d$
$d \not< a$	$d \not< b$	$d \not< c$	$d \not< d$

As the reader may check directly, this relation was constructed specifically to satisfy the definition of a strict total order relation. Totally ordered sets are best visually represented

by having their elements arranged in a line. For example, see the representation of A with $<$ below.

$$a < b < c < d$$

Figure 1.1: Visual representation of the total ordering relation on A .

Partial Order Relations

Here we introduce a generalization of total order relations by removing the property of (strict) comparability.

Definition. A **strict partial order relation** on A is a relation $<$ with the following properties for all $a, b, c \in A$:

- (P1) IRREFLEXIVITY: $a \not< a$,
- (P2) ASYMMETRY: if $a < b$ then $b \not< a$,
- (P3) TRANSITIVITY: if $a < b$ and $b < c$, then $a < c$.

A **weak partial order relation** on A is a relation $\leq$ with the following properties for all $a, b, c \in A$:

- (P1') REFLEXIVITY: $a \leq a$,
- (P2') ANTISYMMETRY: if $a \leq b$ and $b \leq a$, then $a = b$,
- (P3') TRANSITIVITY: if $a \leq b$ and $b \leq c$, then $a \leq c$.

Given $<$ we may define a corresponding $\leq$ and vice-versa by declaring $\forall x \forall y (x \leq y \iff x < y \vee x = y)$ and $\forall x \forall y (x < y \iff x \leq y \wedge x \neq y)$.

If A has a strict or weak partial order relation, then we say A is a **partially ordered set** and introduce it as a pair $\langle A, < \rangle$. If A has a strict $<$ and a corresponding weak $\leq$, we write $b > a$ to denote $a < b$, and $b \geq a$ to denote $a \leq b$.

Example 1.15. Let $A = \{a, b, c, d, e, f\}$ where a, b, c, d, e, f are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b \rangle \vee x = \langle a, c \rangle \vee x = \langle a, d \rangle \vee x = \langle a, e \rangle \vee x = \langle c, d \rangle \vee x = \langle c, e \rangle \vee x = \langle f, b \rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_{<} = \{x \in A \times_K A \mid P(x)\}$, and we form $< = \langle G_{<}, \langle A, A \rangle \rangle$. Since $G_{<} \subseteq A \times_K A$, this is a relation. This gives us the following list:

$$a < b, \quad a < c, \quad a < d, \quad a < e, \quad c < d, \quad c < e, \quad f < b,$$

and $x \not\leq y$ for all cases not included above. As the reader may check directly, this relation was constructed specifically to satisfy the definition of a strict partial order relation.

Whereas total order relations are best represented by arranging elements in a line, partial order relations are best represented by arranging elements in a two-dimensional diagram called a **Hasse diagram**. In such a diagram, if x is below y and there is a path by line segments from x to y , then $x < y$. See the representation of $\langle A, < \rangle$ below.

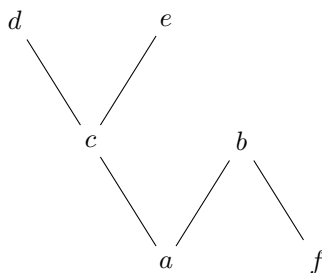


Figure 1.2: Hasse diagram of the partial ordering relation on A .

Below we present some terminology pertaining to partial orders and thus to total orders.

Definition. Let A be a partially ordered set with $<$ as the strict relation and $\leq$ as the corresponding weak relation, and let X be a subset of A . We define the notion of bounds.

- If $b \in A$ such that $\forall x \in X (x \leq b)$, then we say X is **bounded above** and b is an **upper bound** of X .
- If $b \in A$ such that $\forall x \in X (b \leq x)$, then we say X is **bounded below** and b is an **lower bound** of X .

We define the notions of maxima and minima.

- If $m \in X$ such that $\forall x \in X (x \leq m)$, then m is called the **maximum** or **greatest** element of X . If m exists, we write $m = \max X$.
- If $m \in X$ such that $\forall x \in X (m \leq x)$, then m is called the **minimum** or **least** element of X . If m exists, we write $m = \min X$.

We define notations of maximal and minimal elements.

- If $m \in X$ such that $\forall x \in X (m \leq x \implies m = x)$, then m is called a **maximal** element of X .
- If $m \in X$ such that $\forall x \in X (x \leq m \implies m = x)$, then m is called a **minimal** element of X .

In a totally ordered set, the greatest and maximal elements, if they exist, are the same thing. Likewise, in a totally ordered set, the least and minimal elements, if they exist, are the same thing. However, the concepts are not equivalent in general.

Example 1.16. As a simple example, consider $A = \{a, b, c\}$ with only the relationships $a < b$ and $a < c$. Here b and c are maximal, but neither b nor c is the greatest, because they are incomparable with one another.

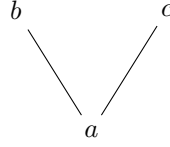


Figure 1.3: Hasse diagram of $\langle A, < \rangle$.

Proposition 1.17. Let $\langle A, < \rangle$ be a partially ordered set. If $a, b \in A$ are the greatest elements of A , then $a = b$. If $c, d \in A$ are the least elements of A , then $c = d$.

Proof. Suppose $a, b \in A$ are the greatest elements of A . By definition of greatest, $b \leq a$ and $a \leq b$, so $a = b$ by (P2'). Suppose $c, d \in A$ are the least elements of A . By definition of least, $c \leq d$ and $d \leq c$, so $c = d$ by (P2'). ■

Let $\langle A, < \rangle$ be a partially ordered set. Proposition 1.17 shows that if the greatest element of A exists, it is unique. In contrast, maximal elements do not have to be unique, as shown in Example 1.16. Likewise, if the least element of A exists, it is unique, unlike minimal elements which do not have to be unique.

Definition. A partially ordered set $\langle A, < \rangle$ is said to be **well-ordered** if every subset of A has a least element. In such a case, we say $<$ is a **well-ordering** on A .

In the closing problem set, we ask the reader to show that every well-ordered partial order is in fact a total order.

Equivalence Relations

Here we introduce another kind of relation.

Definition. An **equivalence relation** on A is a relation $\sim$ with the following properties for all $a, b, c \in A$:

- (E1) REFLEXIVITY: $a \sim a$,
- (E2) SYMMETRY: if $a \sim b$ then $b \sim a$,
- (E3) TRANSITIVITY: if $a \sim b$ and $b \sim c$, then $a \sim c$.

Example 1.18. Let $A = \{a, b, c, d, e\}$ where a, b, c, d, e are distinct elements. Define a relation $\sim$ on A by the following list:

$$a \sim a, \quad a \sim b, \quad a \sim c, \quad b \sim a, \quad b \sim b, \quad b \sim c, \quad c \sim a, \quad c \sim b, \quad c \sim c,$$

$$d \sim d, \quad d \sim e, \quad e \sim d, \quad e \sim e,$$

and $x \not\sim y$ for all cases not included above. Justifying the existence of this relation follows the same process as in previous examples. We may verify this is an equivalence relation by checking all instances where $x \sim y$ and $x \not\sim y$ manually.

Example 1.19. Given any set A , use the existence of $A \times_K A$ and separation to obtain the set $G_E = \{\langle x, y \rangle \in A \times_K A \mid x = y\}$, and then form $E = \langle G_E, \langle A, A \rangle \rangle$. For all $x, y \in A$, we have $x E y$ if and only if $x = y$, so we have internalized the equality relation into a set-coded relation on A . It is easy to see E is an equivalence relation, and in this sense, equivalence relations are a generalization of the equality relation.

Sometimes we want to group various elements of a set A together. We do this as follows.

Definition. A **partition** of a set A is a collection $\mathcal{U}$ of subsets of A such that the following properties hold:

- (a) No member of $\mathcal{U}$ is the empty set: $\forall x \in \mathcal{U} (x \neq \emptyset)$.
- (b) The union of all members of $\mathcal{U}$ is A : $\bigcup \mathcal{U} = A$.
- (c) The intersection of any two distinct members of $\mathcal{U}$ is the empty set: $\forall x, y \in \mathcal{U} (x \neq y \implies x \cap y = \emptyset)$.

Given an equivalence relation $\sim$ on set A , we can form a partition of A as follows. For each $a \in A$, use separation to obtain

$$\{b \in A \mid a \sim b\},$$

which we denote as $[a]$. Then use separation to obtain the set

$$\{x \in \mathcal{P}(A) \mid x = [a] \text{ for some } a \in A\},$$

which we will denote as $A/\sim$. We call $A/\sim$ the **quotient of A by $\sim$** and we refer to elements of $A/\sim$ as **equivalence classes**. We leave it to the reader to show that for any $a, b \in A$ we have $a \sim b$ if and only if $[a] = [b]$, and that $A/\sim$ is a partition of A .

We may also go in the opposite direction. If $\mathcal{U}$ is a partition of A , we may define a relation $\sim$ on A by $a \sim b$ if and only if there exists a common class $C \in \mathcal{U}$ such that $a, b \in C$. We again leave it to the reader to show that the resulting relation is an equivalence relation.

Example 1.20. The quotient of $A = \{a, b, c, d, e\}$ by $\sim$ defined in [Example 1.18](#) is

$$A/\sim = \{\{a, b, c\}, \{d, e\}\}.$$

This is a partition of A into subsets.

1.4. Set-Coded Functions

Basic Definitions

Based on the axioms of what is allowed and not allowed in set theory, we introduced various function-like operations, some of which are defined on the class of all sets like the power set operation $\mathcal{P}$ and the union operation $\bigcup$. Some other operations are defined only if certain non-trivial conditions are satisfied. For example, when it comes to the intersection operation $\bigcap$, we only consider $\bigcap A$ well-formed if there exists a proof that A is nonempty.

Just like with relations, we want to define functions only on certain sets, and we want functions to be treated as objects of set theory. For this reason, we will introduce new functions by encoding them as sets.

Definition. A **(set-coded) function** f from A to B is a relation $f = \langle G_f, \langle A, B \rangle \rangle$ in which the following two properties are satisfied:

- (F1) EXISTENCE/TOTALITY: For each $x \in A$, there exists $y \in B$ such that $\langle x, y \rangle \in G_f$.
- (F2) UNIQUENESS: For all $x \in A$ and $y, y' \in B$, if $\langle x, y \rangle, \langle x, y' \rangle \in G_f$ then $y = y'$.

If f is a function, we call A the **domain**, B the **codomain**, and G_f the **graph**, and we denote it by $f : A \rightarrow B$.

In addition to (set-coded) function, we have the following objects that have weakened conditions.

Definition. A **partial function** f from A to B is a relation $f = \langle G_f, \langle A, B \rangle \rangle$ satisfying (F2) but not necessarily (F1). We say A is the **ambient domain**, B is the **codomain**, G_f is the **graph**, and $S = \{x \in A \mid \exists y \in B : \langle x, y \rangle \in G_f\}$ is the **active domain**. We denote it by $f : A \rightharpoonup B$.

Definition. A **multivalued function** f from A to B is a relation $f = \langle G_f, \langle A, B \rangle \rangle$ satisfying (F1) but not necessarily (F2). We say A is the **domain**, B is the **codomain**, and G_f is the **graph**. We do not denote it by any special notation.

In our setup of terminology, a (set-coded) function is both a partial function and a multivalued function, but partial functions and multivalued functions are not necessarily functions. We will oftentimes refer to a function as a **map** or **mapping**.

We think of the partial function as “sending” elements of the active domain to elements of the codomain. Given $f : A \rightharpoonup B$ with active domain S and $x \in S$, we want a way to automatically denote the unique $y \in B$ for which $\langle x, y \rangle \in G_f$. To this end, let D be the predicate

$$D(f, x) \equiv (f \text{ is a partial function}) \wedge \exists y (\langle x, y \rangle \in \text{Graph}(f))$$

where $\text{Graph}(f)$ is the graph of f , and then note that if f, x satisfy D then there exists a unique y such that $\langle x, y \rangle \in \text{Graph}(f)$. Given this, we define a function symbol App with

domain condition D such that $\text{App}(f, x)$ is the unique y for which $\langle x, y \rangle \in \text{Graph}(f)$. We abbreviate all well-formed terms of the form $\text{App}(f, x)$ as simply $f(x)$. We may sometimes signify the fact that $y = f(x)$ by writing

$$x \mapsto y, \quad f : x \mapsto y, \quad x \xrightarrow{f} y,$$

or some variation thereof.

Example 1.21. Let $A = \{a, b, c, d\}$ and $B = \{x, y, z\}$ where a, b, c, d are distinct elements of A and x, y, z are distinct elements of B . Let P be the predicate

$$P(t) \equiv t = \langle a, y \rangle \vee t = \langle b, x \rangle \vee t = \langle c, x \rangle \vee t = \langle d, x \rangle.$$

By the existence of $A \times_K B$ and separation, we obtain the set $G_f = \{t \in A \times_K B \mid P(t)\}$ and we form $f = \langle G_f, \langle A, B \rangle \rangle$.

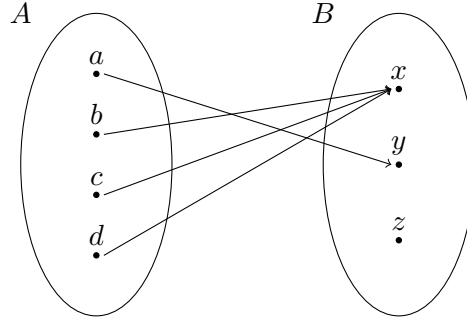


Figure 1.4: Visual representation of the function $f : A \rightarrow B$.

As one may check, properties (F1) and (F2) hold, so f is a function.

Example 1.22. Let $A = \{a, b, c, d\}$ and $B = \mathcal{P}(A)$ where a, b, c, d are distinct elements. Let P be the predicate

$$P(t) \equiv \exists x (x \in A \wedge t = \langle x, \{x\} \rangle).$$

By the existence of $A \times_K B$ and separation, we obtain the set $G_f = \{t \in A \times_K B \mid P(t)\}$ and we form $f = \langle G_f, \langle A, B \rangle \rangle$. As one may check, properties (F1) and (F2) hold, so f is a function.

Below we define some important sets that are associated with partial functions (which includes functions).

Definition. Let $f : A \rightarrow B$ be a partial function and let S be its active domain. We define the **image** of f to be

$$\text{Im } f = \{y \in B \mid y = f(x) \text{ for some } x \in S\}.$$

If $E \subseteq A$, we define the **image of E under f** to be

$$f(E) = \{y \in B \mid y = f(x) \text{ for some } x \in E \cap S\}.$$

If $F \subseteq B$, we define the **inverse image of F under f** to be

$$f^{-1}(F) = \{x \in S \mid f(x) \in F\}.$$

Constructions justifying existence and uniqueness of these is done below.

For any partial function, let $\text{AmbDom}(f)$ be the ambient domain of f , let $\text{ActDom}(f)$ be the active domain of f , and let $\text{Cod}(f)$ be the codomain of f . In the case where f is a function, the active and ambient domains are the same and we just call them the domain $\text{Dom}(f)$. For any pair $c = \langle a, b \rangle$, let us write $\pi_L(c) = a$ and $\pi_R(c) = b$. Define the predicate

$$P(y, c) \equiv (c \text{ is a Kuratowski ordered pair}) \wedge (\pi_L(c) \text{ is a partial function}) \\ \wedge \pi_R(c) \subseteq \text{AmbDom}(\pi_L(c)) \wedge \exists x (x \in \pi_R(c) \cap \text{ActDom}(\pi_L(c)) \wedge y = \text{App}(\pi_L(c), x)),$$

and let $D(f, E)$ say f is a partial function and $E \subseteq \text{AmbDom}(f)$. Then for all f, E such that $D(f, E)$ holds, by separation there exists a unique set of the form

$$\text{AppIm}(f, E) = \text{Sep}_P(\text{Cod}(f), \langle f, E \rangle),$$

where $\text{Sep}_P(A, c)$ is the subset of A of elements y satisfying $P(y, c)$, so AppIm is defined for all f, E for which $D(f, E)$ holds. Now $\text{AppIm}(f, E)$ is precisely the image of E under f , and we abbreviate it as $f(E)$ as in the above definition. The image of f is now simply $\text{Im } f = \text{AppIm}(f, \text{AmbDom}(f))$. The existence and uniqueness of inverse images can be shown similarly.

Example 1.23. Recall the function from [Example 1.21](#) where $A = \{a, b, c, d\}$, $B = \{x, y, z\}$, and $f : A \rightarrow B$ is defined by $f(a) = y$, $f(b) = x$, $f(c) = x$, and $f(d) = x$. The codomain is $B = \{x, y, z\}$, but the image is $\text{Im } f = \{x, y\}$.

Example 1.24. Recall the function from [Example 1.22](#) where $A = \{a, b, c, d\}$, $B = \mathcal{P}(A)$, and $f : A \rightarrow B$ is defined by $f(x) = \{x\}$. The codomain is $\mathcal{P}(A)$, but the image is $\text{Im } f = \{\{a\}, \{b\}, \{c\}, \{d\}\}$.

Given two sets, we may consider all possible functions from one to the other.

Definition. Given sets A and B , the set of all functions $f : A \rightarrow B$ from A to B is denoted by B^A or $\text{Map}(A, B)$.

Let us justify the existence of $\text{Map}(A, B)$ given sets A, B .

Proposition 1.25. *Given sets A and B , the set of all functions from A to B exists and is unique.*

Proof. First, we collect all graphs of functions from A to B into a set as follows. By taking the power set, we obtain $\mathcal{P}(A \times_K B)$. This is the collection of all sets $G \subseteq A \times_K B$. Let P be the predicate

$$P(G, c) \equiv (c \text{ is a Kuratowski ordered pair}) \wedge G \subseteq \pi_L(c) \times_K \pi_R(c) \\ \wedge (G \text{ satisfies (F1) and (F2) with respect to } \pi_L(c) \times_K \pi_R(c)).$$

By separation, there exists a set

$$F = \{G \in \mathcal{P}(A \times_K B) \mid P(G, \langle A, B \rangle)\}.$$

This is the set of graphs of functions from A to B .

To proceed, we invoke the axiom schema of replacement (ZFC7) to attach the domain and codomain to the graphs. Let Q be the predicate

$$Q(G, f, c) \equiv f = \langle G, c \rangle.$$

Let us check that this is a functional predicate. Given G and c , we may form $f = \langle G, c \rangle$. If $Q(G, f, c)$ and $Q(G, \tilde{f}, c)$, then $f, \tilde{f}$ are both $\langle G, c \rangle$, so $f = \tilde{f}$. Thus, Q is a functional predicate and by the axiom schema of replacement (ZFC7) with parameter $c = \langle A, B \rangle$, there exists a set $\tilde{F} = \text{Rep}_Q(F, \langle A, B \rangle)$ such that the following holds:

- (i) If $f \in \tilde{F}$, then there exists $G \in F$ such that $Q(G, f, \langle A, B \rangle)$, so $f = \langle G, \langle A, B \rangle \rangle$.
- (ii) If $G \in F$, then there exists $f \in \tilde{F}$ such that $Q(G, f, \langle A, B \rangle)$, so $f = \langle G, \langle A, B \rangle \rangle$.

So, given A, B , we took

$$\tilde{F} = \text{Rep}_Q(\text{Sep}_P(\mathcal{P}(A \times_K B), \langle A, B \rangle), \langle A, B \rangle).$$

Next, we verify $\tilde{F}$ is the set of all maps from A to B as intended. Suppose f is a function from A to B . By definition, $f = \langle G_f, \langle A, B \rangle \rangle$ such that $G_f \subseteq A \times_K B$ satisfies (F1) and (F2). We find $G_f \in F$, so by property (ii), there exists some $\tilde{f} \in \tilde{F}$ such that

$$\tilde{f} = \langle G_f, \langle A, B \rangle \rangle.$$

The right-hand side is precisely f , so $f \in \tilde{F}$. Conversely, suppose $f \in \tilde{F}$. By property (i), there exists some $G \in F$ such that $f = \langle G, \langle A, B \rangle \rangle$. Given that $G \in F$, G satisfies (F1) and (F2), so f is a function.

Thus, f is a function from A to B if and only if $f \in \tilde{F}$. The uniqueness of the set of all functions from A to B follows straightforwardly by extensionality. $\blacksquare$

Compositions and Changes of Domain and Codomain

We will be using the following useful lemma.

Lemma 1.26. *Suppose $f, g : A \rightarrow B$ are functions with the same domain A and codomain B . If $f(x) = g(x)$ for all $x \in A$, then $f = g$.*

Proof. Write $f = \langle G_f, \langle A, B \rangle \rangle$ and $g = \langle G_g, \langle A, B \rangle \rangle$. We aim to show the graphs G_f, G_g are equal. Given $t \in G_f$, we know $t = \langle x, f(x) \rangle$ for some $x \in A$. Since $f(x) = g(x)$ by hypothesis, substitution implies $t = \langle x, g(x) \rangle$, and $\langle x, g(x) \rangle \in G_g$. Thus, by substitution, $t \in G_g$. Conversely, if $t \in G_g$, analogous reasoning shows $t \in G_f$. By extensionality, $G_f = G_g$ and so $f = g$. ■

We introduce composition of functions.

Definition (Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. We define the **composition** of these two functions to be the function $g \circ f : A \rightarrow C$ such that $g \circ f(a) = g(f(a))$ for all $a \in A$. We verify $g \circ f$ exists and is unique below.

Proposition 1.27 (Existence and uniqueness of compositions). *If $f : A \rightarrow B$ and $g : B \rightarrow C$ are functions, then the composition $g \circ f$ exists and is unique.*

Proof. Write $f = \langle G_f, \langle A, B \rangle \rangle$ and $g = \langle G_g, \langle B, C \rangle \rangle$. By the existence of $A \times_K C$ and by separation, there exists the set

$$G = \{t \in A \times_K C \mid P(t, \langle f, g \rangle)\}$$

where

$$\begin{aligned} P(t, c) &\equiv (c \text{ is a Kuratowski ordered pair}) \wedge (\pi_L(c), \pi_R(c) \text{ are functions}) \\ &\quad \wedge \text{Cod}(\pi_L(c)) = \text{Dom}(\pi_R(c)) \\ &\quad \wedge \exists x (x \in \text{Dom}(\pi_L(c)) \wedge t = \langle x, \text{App}(\pi_R(c), \text{App}(\pi_L(c), x)) \rangle). \end{aligned}$$

Now define $h = \langle G, \langle A, C \rangle \rangle$, which exists. Since $G \subseteq A \times_K C$, it remains to verify (F1) and (F2).

For (F1), for each $x \in A$, we have $\langle x, g(f(x)) \rangle \in A \times_K C$ and $P(\langle x, g(f(x)) \rangle, \langle f, g \rangle)$, so

$$\langle x, g(f(x)) \rangle \in G,$$

so (F1) holds. For (F2), suppose $x \in A$ and $z, z' \in C$ such that

$$\langle x, z \rangle, \langle x, z' \rangle \in G.$$

By definition of G , there exist $u, u' \in A$ such that

$$\langle x, z \rangle = \langle u, g(f(u)) \rangle \quad \text{and} \quad \langle x, z' \rangle = \langle u', g(f(u')) \rangle.$$

By (1.1), $x = u$, $z = g(f(u))$, $x = u'$, and $z' = g(f(u'))$. As a result we have $u = u'$, so by substitution, $g(f(u)) = g(f(u'))$, and so $z = z'$ and (F2) holds.

This shows $h : A \rightarrow C$ is a function such that $h(x) = g(f(x))$ for all $x \in A$ by construction. Uniqueness follows from [Lemma 1.26](#). ■

By existence and uniqueness of $g \circ f$ given composable functions f, g , we may create a composition formation operator as follows. Take the domain condition D and predicate P so that

$$D(f, g) \equiv (f, g \text{ are functions}) \wedge \text{Cod}(f) = \text{Dom}(g)$$

and

$$\begin{aligned} P(f, g, h) \equiv & D(f, g) \wedge (h \text{ is a function}) \wedge \text{Dom}(h) = \text{Dom}(f) \\ & \wedge \text{Cod}(h) = \text{Cod}(g) \wedge \forall x \in \text{Dom}(f) (h(x) = g(f(x))). \end{aligned}$$

Given

$$\forall f \forall g (D(f, g) \implies \exists! h P(f, g, h)),$$

we obtain the operation Comp with domain condition D such that $h = \text{Comp}(f, g)$ is the composition. We abbreviate $\text{Comp}(f, g)$ as $g \circ f$.

We introduce changes of domains and codomains of functions.

Definition (Change of domain and codomain). Let $f : A \rightarrow B$, $X \subseteq A$, and $Y \supseteq f(X)$. We define $f|_X^Y$ to be the function from X to Y such that $f|_X^Y(x) = f(x)$ for all $x \in X$. Notation for special cases:

- If $Y = B$ (the codomain isn't changed), we write $f|_X^B = f|_X$.
- If $X = A$ (the domain isn't changed), we write $f|_A^Y = f|_A^Y$.

We verify $f|_X^Y$ exists and is unique below.

Proposition 1.28 (Existence and uniqueness of change of domain and codomain). *If $f : A \rightarrow B$ is a function, $X \subseteq A$, and $Y \supseteq f(X)$, then the function with a change of domain and codomain $f|_X^Y$ exists and is unique.*

Proof. Given $f = \langle G_f, \langle A, B \rangle \rangle$, $X \subseteq A$, and $Y \supseteq f(X)$, use separation to obtain the subset

$$G = \{t \in G_f \mid P(t, \langle X, Y \rangle)\}$$

where

$$P(t, c) \equiv (c \text{ is a Kuratowski ordered pair}) \wedge t \in \pi_L(c) \times_K \pi_R(c),$$

and then form $h = \langle G, \langle X, Y \rangle \rangle$. If $t \in G$, then $P(t, \langle X, Y \rangle)$, so $t \in X \times_K Y$. Hence $G \subseteq X \times_K Y$, and it remains to verify (F1) and (F2).

For (F1), for each $x \in X$, we have $\langle x, f(x) \rangle \in G_f$ and $f(x) \in f(X) \subseteq Y$, and so $P(\langle x, f(x) \rangle, \langle X, Y \rangle)$. Thus,

$$\langle x, f(x) \rangle \in G,$$

so (F1) holds. For (F2), suppose $x \in X$ and $y, y' \in Y$ such that

$$\langle x, y \rangle, \langle x, y' \rangle \in G.$$

Since $G \subseteq G_f$ and G_f has property (F2), it follows $y = y'$, and so (F2) holds.

This shows $h : X \rightarrow Y$ is a function such that $h(x) = f(x)$ for all $x \in X$ by construction. Uniqueness follows from [Lemma 1.26](#). ■

By existence and uniqueness of $f|_X^Y$ given $f : A \rightarrow B$, $X \subseteq A$, and $Y \supseteq f(X)$, we may create a domain/codomain change operator as follows. Take the domain condition D and predicate P so that

$$\begin{aligned} D(f, p) &\equiv (f \text{ is a function}) \wedge (p \text{ is a Kuratowski ordered pair}) \\ &\quad \wedge \pi_L(p) \subseteq \text{Dom}(f) \wedge \pi_R(p) \supseteq f(\pi_L(p)) \end{aligned}$$

and

$$\begin{aligned} P(f, p, h) &\equiv D(f, p) \wedge (h \text{ is a function}) \wedge \text{Dom}(h) = \pi_L(p) \\ &\quad \wedge \text{Cod}(h) = \pi_R(p) \wedge \forall x \in \pi_L(p) (f(x) = h(x)). \end{aligned}$$

Given

$$\forall f \forall p (D(f, p) \implies \exists! h P(f, p, h)),$$

we obtain the operation Change with domain condition D such that $h = \text{Change}(f, p)$ is the function obtained from f by restricting the domain to $X = \pi_L(p)$ and changing the codomain to $Y = \pi_R(p)$. We abbreviate $\text{Change}(f, \langle X, Y \rangle)$ as $f|_X^Y$.

Identity Maps and Inverses

Now we present some ubiquitous types of functions.

Definition (The identity map). Given a set A , we define the **identity map** of A to be the function denoted $\text{id}_A : A \rightarrow A$ that outputs exactly the same element that is given as the input: $\text{id}_A(x) = x$ for all $x \in A$.

We leave it to the reader to construct the identity map and prove uniqueness in the ending problem set.

Definition. Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be maps.

- g is said to be a **left-inverse** of f if $g \circ f = \text{id}_A$.
- g is said to be a **right-inverse** of f if $f \circ g = \text{id}_B$.
- g is said to be an **inverse** of f if it is both a left-inverse and right-inverse. We leave it to the reader to show that if it exists, then it is unique. We denote it by f^{-1} , which can be done due to uniqueness.

Example 1.29 (Left-inverse). Let $A = \{a, b, c\}$ and $B = \{x, y, z, w\}$ where a, b, c, x, y, z, w are distinct elements. Define $f : A \rightarrow B$ by $f(a) = x$, $f(b) = y$, and $f(c) = z$. Define $g : B \rightarrow A$ by $g(x) = a$, $g(y) = b$, $g(z) = c$, and $g(w) = c$. Then

$$g(f(a)) = a, \quad g(f(b)) = b, \quad \text{and} \quad g(f(c)) = c,$$

so $g \circ f = \text{id}_A$ and g is a left-inverse of f . However, $f(g(w)) = z \neq w$, so g is not a right-inverse of f .

Example 1.30 (Right-inverse). Let $A = \{a, b, c, d\}$ and $B = \{x, y, z\}$ where a, b, c, d, x, y, z are distinct elements. Define $f : A \rightarrow B$ by $f(a) = x$, $f(b) = y$, $f(c) = z$, and $f(d) = z$. Define $g : B \rightarrow A$ by $g(x) = a$, $g(y) = b$, and $g(z) = c$. Then

$$f(g(x)) = x, \quad f(g(y)) = y, \quad \text{and} \quad f(g(z)) = z,$$

so $f \circ g = \text{id}_B$ and g is a right-inverse of f . However, $g(f(d)) = c \neq d$, so g is not a left-inverse of f .

Example 1.31 (Inverse). Let $A = \{a, b, c\}$ and $B = \{x, y, z\}$ where a, b, c, x, y, z are distinct elements. Define $f : A \rightarrow B$ by $f(a) = y$, $f(b) = z$, and $f(c) = x$. Define $g : B \rightarrow A$ by $g(x) = c$, $g(y) = a$, and $g(z) = b$. Then

$$g(f(a)) = a, \quad g(f(b)) = b, \quad \text{and} \quad g(f(c)) = c,$$

so $g \circ f = \text{id}_A$. Also,

$$f(g(x)) = x, \quad f(g(y)) = y, \quad \text{and} \quad f(g(z)) = z,$$

so $f \circ g = \text{id}_B$. Thus, g is an inverse of f .

Injective, Surjective, and Bijective Maps

Appendices

